

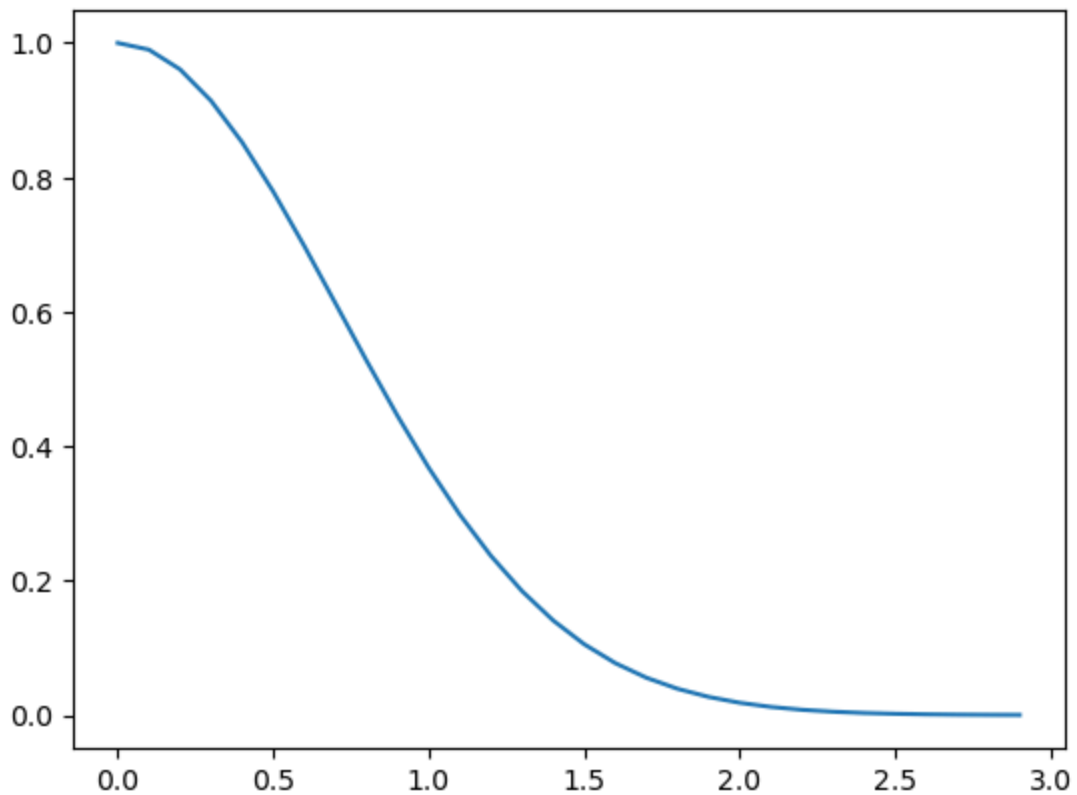
5.3

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [9]: def g(t):
        return np.exp(-(t**2))
t = np.arange(0,3,0.1)

fig, ax = plt.subplots()
ax.plot(t, g(t))
```

Out[9]: [



```
In [10]: def simp1(func, a, b, steps):
        if steps%2!=0:
            steps = steps + 1
            print('i added one step')
        dx = (b-a)/steps
        x = np.linspace(a, b, steps+1)
        y = func(x)
        s = dx/3*(y[0]+y[-1])
        s = s + dx/3*(4*np.sum(y[1:steps:2]) + 2*np.sum(y[2:steps-1:2]))
        return(s)
```

```

def f(x):
    def g(t):
        return(np.exp(-(t**2)))
    integral = 0.0
    integral = simpl(g, 0, x, 1000)
    return(integral)

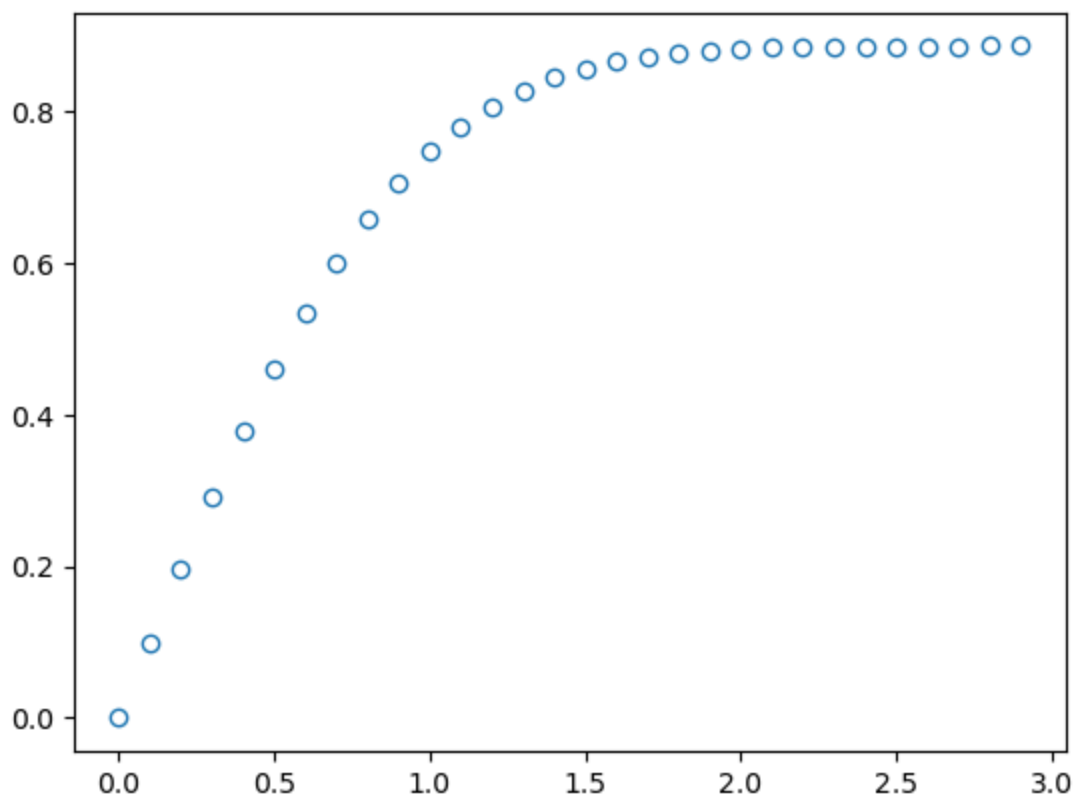
x = np.arange(0,3,0.1)
y = []
for i in x:
    y.append(f(i))

fig, ax = plt.subplots()

ax.plot(x, y, 'o', mfc='none')

```

Out[10]: [



In []:

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In []:

In [5]: `x = np.arange(0,3.1,0.1)`

In [6]: x

Out[6]: array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. , 1.1, 1.2,
1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2. , 2.1, 2.2, 2.3, 2.4, 2.5,
2.6, 2.7, 2.8, 2.9, 3.])

In []: